

Day 5: Configuring OSPF Practical

Task 1 :- Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

ANS :-

1.Place and Connect Devices :

- Place 3 Routers (R1, R2, R3).
- Connect R1 to R2 using a serial or gigabit cable.
- Connect R2 to R3 using a serial or gigabit cable.

2.Assign IP Addresses :

- R1 - R2 Link (Area 0) :
 - R1 interface: 10.1.1.1 /24
 - R2 interface: 10.1.1.2 /24
- R2 - R3 Link (Area 1) :
 - R2 interface: 10.2.2.1 /24
 - R3 interface: 10.2.2.2 /24
- Turn ON all interfaces (no shutdown).

3.Configure OSPF on Routers :

R1 (Area 0 only) :

```
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
```

R2 (ABR - Connects Area 0 and Area 1) :

```
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 1
```

R3 (Area 1 only) :

```
router ospf 1
network 10.2.2.0 0.0.0.255 area 1
```

4.Verify Connectivity :

- Ping from R1 to R3 (ping 10.2.2.2).
- Run show ip route on R1 to confirm OSPF routes are learned.

Task 2 :- Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.

ANS :-

Router 1 (Internal - Area 0) :-

```
Router1(config)# router ospf 1
Router1(config-router)# network 10.1.1.0 0.0.0.255 area 0
```

Router 2 (Area Border Router / ABR) :-

Connects both Area 0 and Area 1.

```
Router2(config)# router ospf 1
Router2(config-router)# network 10.1.1.0 0.0.0.255 area 0
Router2(config-router)# network 192.168.1.0 0.0.0.255 area 1
```

Router 3 (Internal - Area 1) :-

```
Router3(config)# router ospf 1
Router3(config-router)# network 192.168.1.0 0.0.0.255 area 1
```

Verify Connection :-

Run this on any router to check if neighbors are connected:

```
Router# show ip ospf neighbor
```

Look for: State should say FULL (meaning routers are talking and fully synced).

Task 3 :- Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other..

ANS :-

1.Connect the PCs:

Connect PC1 to your Area 0 switch or router, and PC2 to your Area 1 switch or router using Ethernet cables.

2.Configure IP Addresses:

Set the IP settings on both computers:

- PC1 (Area 0): IP 192.168.10.2 | Subnet 255.255.255.0 | Gateway 192.168.10.1
- PC2 (Area 1): IP 192.168.20.2 | Subnet 255.255.255.0 | Gateway 192.168.20.1

3.Advertise Networks in OSPF:

Tell each router about its connected PC subnet:

- Area 0 Router: network 192.168.10.0 0.0.0.255 area 0
- Area 1 Router: network 192.168.20.0 0.0.0.255 area 1

4.Ping to Test:

Open Command Prompt on PC1 and run:

```
ping 192.168.20.2
```

Receiving reply messages confirms your two areas are communicating.

Task 4 :- Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface.

ANS :-

1.Shut Down the Interface :

Log into your ABR (Area Border Router) and turn off the link connecting Area 0 and Area 1 :

```
interface GigabitEthernet0/0  
shutdown
```

2.Observe Network Failure :

Run these commands to confirm OSPF lost the connection :

- show ip route — Routes to Area 1 disappear from the table.
- show ip ospf neighbor — The neighbor state drops to DOWN.

3.Re-enable the Interface :

Turn the interface back on :

```
interface GigabitEthernet0/0  
no shutdown
```

4.Verify OSPF Restoration :

Check that OSPF reconverges and restores the path:

- show ip ospf neighbor — State goes back to FULL.
- show ip route — Area 1 routes show up again.
- ping 192.168.20.2 — Ping responds successfully.